



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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(DATABASE) - (SECP2523)

DDL & DML

TITLE : REPORTING MODULE

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DDL

The reporting module of the system utilizes the already existing tables of the database in generating reports. Instead of changing the structure of the database (DDL operations), it uses DML operations to fetch and update data. Listed below are the key arguments in favor of such an approach:

1. Data Aggregation without structural alteration

Reporting entails the aggregation of data based on month and year for summaries. The control logic will be responsible for this through SELECT queries along with aggregations and computations. There will not be any DDLs, since there are no new tables, columns, or any other modification in the database schema.

2. Current Sources of Data

There is no requirement to create new database objects or modify existing ones since all reports utilize data from tables that are already in place. The module only requires data retrieval (SELECT) and, in certain cases, exporting the data to physical files (PDF); these functions are handled outside of the database layer.

3. Security and User Input Considerations

The reporting module implements specific security measures or restrictions through authentication, since reports are generated dynamically based on user input. Its primary purpose is to gather and present data to administrative users in a clear format. Any necessary computations or data manipulations for the report are handled by the controller, rather than the database structure itself.

In conclusion, DML is sufficient for implementing the reporting module, as it only involves read operations concerning the database structure. There is no requirement to include DDL commands in this part of the project since no schema changes are necessary.

DML

- **DML for membership_notif**

This DML is to show the number of applicants that apply for membership, their status, and its approval date.

1. By applicationStatus = lulus,

```
1 SELECT applicationID, applicationStatus, approvalDate
2 FROM `membership_notif`
3 WHERE applicationStatus = 'lulus';
```

	applicationID	applicationStatus	approvalDate
☐ Edit ☐ Copy ☐ Delete	2	lulus	2025-01-14
☐ Edit ☐ Copy ☐ Delete	4	lulus	2025-01-14
☐ Edit ☐ Copy ☐ Delete	5	lulus	2025-01-14

2. By applicationStatus = diproses,

```
5 SELECT applicationID, applicationStatus, approvalDate
6 FROM `membership_notif`
7 WHERE applicationStatus = 'diproses';
```

	applicationID	applicationStatus	approvalDate
☐ Edit ☐ Copy ☐ Delete	7	diproses	0000-00-00

3. By approvalDate in year 2025,

```
9 SELECT applicationID, applicationStatus, approvalDate
10 FROM `membership_notif`
11 WHERE YEAR(approvalDate) = 2025;
```

	applicationID	applicationStatus	approvalDate
☐ Edit ☐ Copy ☐ Delete	2	lulus	2025-01-14
☐ Edit ☐ Copy ☐ Delete	4	lulus	2025-01-14
☐ Edit ☐ Copy ☐ Delete	5	lulus	2025-01-14

4. By total,

```
13 SELECT applicationID, applicationStatus, approvalDate
14 FROM `membership_notif`
```

	applicationID	applicationStatus	approvalDate
☐ Edit Copy Delete	2	lulus	2025-01-14
☐ Edit Copy Delete	4	lulus	2025-01-14
☐ Edit Copy Delete	5	lulus	2025-01-14
☐ Edit Copy Delete	7	diproses	0000-00-00

- **DML for stock_record**

This DML is to show the sum of shareCapital (Modal Syer), feeCapital(Modal Yuran), and memberFund(Tabung Anggota) that are accumulated by month and year.

```
1 SELECT shareCapital, feeCapital, memberFund, recordDate
2 FROM `stock_record`
```

	shareCapital	feeCapital	memberFund	recordDate
☐ Edit Copy Delete	300.00	150.00	5.00	2025-01-15

If recordDate = 2024,

```
1 SELECT shareCapital, feeCapital, memberFund, recordDate
2 FROM `stock_record`
3 WHERE YEAR(recordDate) = 2024;
```

shareCapital	feeCapital	memberFund	recordDate
Query results operations			
Create view			

(no information appear, as there's no data from 2024)

- **DML for loan_notif**

This DML is to show the number of loan application for loan type, status of the loan application and total amount of loan that are accumulated by month and year.

```
1 SELECT loanApplicationStatus, approvalDate, loanType, loanAmount
2 FROM `loan_notif`
```

	loanApplicationStatus	approvalDate	loanType	loanAmount
☐ Edit Copy Delete	Sedang Diproses	NULL	al-bai	1000.00

If loanApplicationStatus = lulus,

```
1 SELECT loanApplicationStatus, approvalDate, loanType, loanAmount
2 FROM `loan_notif`
3 WHERE loanApplicationStatus = 'Lulus';
```

loanApplicationStatus	approvalDate	loanType	loanAmount
Query results operations			

(no information appear, as there's no successful status for loan application in data)

If loanType = al-innah,

```
5 SELECT loanApplicationStatus, approvalDate, loanType, loanAmount
6 FROM `loan_notif`
7 WHERE loanType = 'al-innah';
```

loanApplicationStatus	approvalDate	loanType	loanAmount
Query results operations			

(no information appear, as there's no application for loan type al-inna in data)